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ORIGINAL ARTICLE

Psoriasis increases risk of new-onset atrial fibrillation: a systematic review and meta-analysis of prospective observational studies

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ABSTRACT

Background: Psoriasis is a common chronic immune-mediated dermatological disease that increases the risk of cardiovascular disease. We conducted a systematic review and meta-analysis to evaluate the association between psoriasis and atrial fibrillation from prospective observational studies.

Methods: A comprehensive search of the databases of the MEDLINE and EMBASE was performed from inception through November 2015. The inclusion criterion was the prospective observational study that assessed the risk of new-onset atrial fibrillation in adults with psoriasis. Outcome was the adjusted hazard ratio (HR) of atrial fibrillation comparison between patients with psoriasis and controls. Pooled HR and 95% confidence intervals (CI) were calculated using a random-effects model.

Results: The initial search yielded 176 articles. Fifteen articles underwent full-length review and data were extracted from 4 observational studies. Incidence of atrial fibrillation was ascertained by cardiologist-reviewed electrocardiograms. There was a significant increased risk of new-onset atrial fibrillation in patients with psoriasis compared to controls with a pooled HR 1.42 (95%CI 1.22–1.65).

Conclusion: Our meta-analysis of prospective studies demonstrated that patients with psoriasis have increased risk of new-onset atrial fibrillation. Future interventional studies addressing the impact of psoriasis treatment and prevention of atrial fibrillation should be performed.

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KEYWORDS
Psoriasis; atrial fibrillation; meta-analysis

Introduction

Psoriasis is a chronic, immune-mediated systemic inflammatory disease that according to the National Health and Nutrition Examination Survey affects 3.2% of the US population, accounting for 7.2 million adults in the US alone (1). Psoriasis manifests not only with physical manifestations affecting skin and joints, but recently a lot of research has gone into the emotional toll psoriasis takes on a person's psychological well-being. The common comorbidities of cutaneous psoriasis include psoriatic arthritis (PsA), Crohn's disease, uveitis and depression (2,3). In addition, cardiovascular disease risk factors (including metabolic syndrome) are seen more frequently among patients with psoriasis, and strong epidemiologic evidence has demonstrated that psoriasis is independently associated with myocardial infarction. As these comorbid conditions and other medical complications adversely affect morbidity and mortality in patients with psoriasis, dermatologists working with this heinous disease can play an important role in promptly identifying and, when necessary, referring patients for further workup and treatment when signs or symptoms of these comorbidities or complications are observed (4).

Recent studies have brought to light the association between psoriasis and atrial fibrillation (AF) (5), which is the most common cardiac arrhythmia encountered in clinical practice and is in itself associated with increased risk of morbidity and mortality (6). There is recent evidence to support possible implication of oxidative stress and inflammation as important contributors to the pathophysiology of atrial fibrillation (7,8).

The authors of this systemic review and meta-analysis reviewed the current evidence that support a possible relationship between psoriasis and atrial fibrillation by examining previous epidemiologic studies and obtaining a conclusion based on the findings.

Materials and methods

This systematic review and meta-analysis was conducted and reported according to the Meta-analysis Of Observational Studies in Epidemiology statement (9).

Search strategy

Two authors (AS, SU) independently searched published studies indexed in the MEDLINE and EMBASE databases. References of all selected studies were also examined. The following main search terms were used: psoriasis and atrial fibrillation. The full search terms used are detailed in the Appendix.

Inclusion and exclusion criteria

This review included all published prospective observational studies that assessed the association of psoriasis and atrial fibrillation. Reviews, case reports and abstracts were excluded because their quality of studies could not be assessed.

We included studies that recruited participants aged 18 years or older who had psoriasis. The main outcome of this study was new onset atrial fibrillation reported as incidence, odds ratio (OR), risk ratio (RR), or hazard ratio (HR) comparing between patients with psoriasis at baseline and without psoriasis. Mild psoriasis was defined if a patient was treated only with topical medications

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during the follow-up period, whereas moderate-to-severe psoriasis was defined if a patient was treated with phototherapy or systemic medications during the follow-up period.

Data extraction

Two authors independently reviewed titles and abstracts of all citations that were identified. After all abstracts were reviewed, data comparisons between the two investigators were conducted to ensure completeness and reliability. The inclusion criteria were independently applied to all identified studies. Differing decisions were resolved by consensus.

Full-text versions of potentially relevant papers identified in the initial screening were retrieved. If multiple articles from the same study were found, only the article with the most complete data was included to prevent data duplication. Data concerning authors, year of publication, source, study design, baseline participant characteristics, type of psoriasis, duration of psoriasis, assessment of atrial fibrillation and factors used in the multivariate model were independently extracted. We planned to contact the authors of the primary reports to request any unpublished data. If the authors did not reply, we planned to use the available data for our analyses.

Assessment of quality

A subjective assessment of methodological quality for observational studies was evaluated by two authors (AS and SU) using the Newcastle–Ottawa Scale (NOS) (10). The NOS is a quality assessment tool for non-randomized studies. It uses a "star system" based on three major perspectives: selection of the study groups (0–4 stars, or 0–5 stars for cross-sectional study), comparison of the groups by controlling for important and additional relevant factors (0–2 stars) and the ascertainment of outcome of interest or exposure (0–3 stars). A total score of 3 or less was considered poor, 4–6 was considered moderate and 7–9 was deemed high quality. We excluded studies from our meta-analysis if they had poor quality (score $\leq$3). Discrepant opinions between authors were resolved by consensus.

Statistical analysis

We performed meta-analysis of the included studies using Comprehensive Meta-Analysis 3.3 software from Biostat, Inc (Englewood, NJ). We calculated pooled hazard ratio (HR) of outcome for cohort studies in multivariable model. The adjusted factors included age, sex, co-morbidity, smoker, socioeconomic data. All outcomes were reported as effect estimate and its 95% confidence intervals (CIs) comparing between psoriasis and control groups using a random-effects model. We excluded studies from meta-analysis and only presented the result with narrative description (qualitative analysis) when there were not sufficient data available for calculating pooled effect size. The heterogeneity of effect estimates across these studies was quantified using the Q statistic and I^2 ($p < .10$ was considered statistically significant). The Q statistic compared the observed between-study dispersion and expected dispersion of the effect size, and was expressed in p values for statistical significance. An I^2 is the ratio of true heterogeneity to total observed variation. An I^2 of 0–40% was considered to exclude heterogeneity, of 30–60% was considered to represent moderate heterogeneity, of 50–90% was considered to represent substantial heterogeneity, and of 75–100% was considered to represent considerable heterogeneity (11). A sensitivity analysis was performed to assess influence of the individual studies on the overall results by omitting one study at a time and using fixed-effect analysis when $I^2 < 40\%$. Subgroup analysis and meta-regression were performed to find factors that may explain heterogeneity or difference in outcome between each study. Publication bias was assessed using funnel plot, Egger's regression test and its implications with the trim and fill method (12).

Results

Description of included studies

The initial search yielded 176 articles (Appendix); 161 articles were excluded because they were not observational studies (95 articles), did not include patients with psoriasis (34 articles), or did not measure new onset AF as an outcome (32 articles). A total of 15 articles underwent full-length review. Data were extracted from four studies for qualitative analysis (13–16). All studies had prospective designs, involving totally 12,755 participants with a median follow-up time between 2.5 and 5 years. Psoriasis was identified by a medical code associated with clinic visits to a dermatologist. Incidence of atrial fibrillation was ascertained by cardiologist-reviewed electrocardiograms. The characteristics of the four extracted studies included in this review are outlined in Table 1.

Quality assessment of included studies

The quality of the included studies was evaluated by NOS (Table 1). Total score ranged from 5–8. None of the studies was excluded from the meta-analysis by having poor quality.

Meta-analysis results

There was a statistically significant higher risk of new-onset AF in patients who had psoriasis with pooled HR of 1.42 (95%CI 1.22–1.65). The statistical between-study heterogeneity (I^2) was 68%, $p = .01$ (Figure 1). In subgroup analysis by severity of psoriasis (Figure 2), there is no difference between mild (HR =1.22, 95%CI 1.15–1.30) and severe (HR =1.51, 95%CI 1.22–1.87) psoriasis (p value between subgroup was .06).

Publication bias

Publication bias could not be assessed as there are too few included studies.

Discussion

Atrial fibrillation is the most common cardiac arrhythmia and also accounts for more than 25% of all strokes (17). For this reason, it becomes important to identify predisposing factors for atrial fibrillation in order to reduce risk factors and work toward prevention strategies. There is an established association between psoriasis and cardiovascular disease, which supports the hypothesis that patients with psoriasis may also have an increased incidence of atrial fibrillation.

The current study is, to our knowledge, the first systematic and meta-analysis review to examine the risk of AF in psoriatic patients in four studies demonstrating the relationship between new onset atrial fibrillation and psoriasis. For two of these four studies, we did a subgroup analysis of mild and severe psoriasis. So, a total of three analyses were conducted, one combining results of mild and severe psoriasis together, one using only mild psoriasis, and one using severe psoriasis.

Table 1. Characteristics of included studies.

Study (year)	Design	Study period	Study population	Exclusion criteria	Participants (N)	End point	Factors adjusted in multivariable analysis
Ahlehoff (2012)	Prospective cohort	1997–2006	Entire Danish Population	Previous atrial fibrillation or stroke	4,518,484	Atrial fibrillation, ischemic stroke	Calendar year, concomitant medication, co-morbidity, socioeconomic data, and gender
Bang (2014)	Prospective cohort	N/A	Hypertensive patients with left ventricular hypertrophy	Established heart failure, myocardial infarction, or atrial fibrillation	7099	Atrial fibrillation, heart failure	Age, sex, hemoglobin, diabetes, time-varying SBP, heart rate, study treatment and Sokolow–Lyon hypertrophy
Armstrong (2013)	Prospective cohort	2004–2009	Database of University of California Davis	Not on cardiac catheterization list	507	Atrial fibrillation	Race, dyslipidemia, chronic obstructive pulmonary disease, or hypertension
Parisi (2015)	Prospective cohort	1994–2009	Patients aged 20 years or older	History of cardiovascular disease	48,523	Major cardiovascular events	Inflammatory arthritis, diabetes, chronic kidney disease, hypertension, transient ischemic attack, atrial fibrillation, valvular heart disease, thromboembolism, congestive heart failure, depression, current smoker, age, and gender

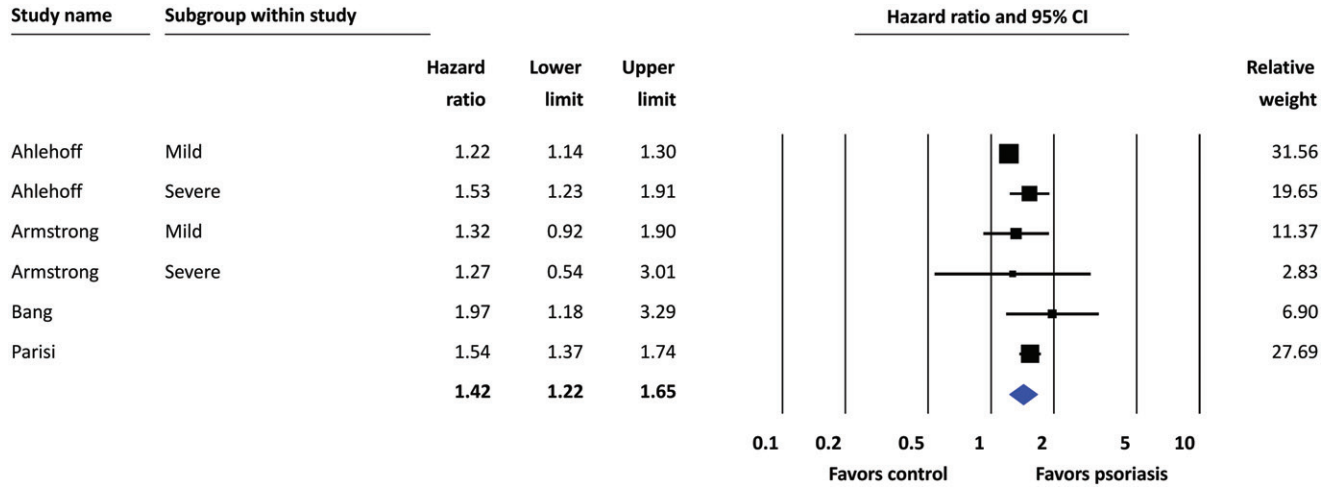

Figure 1. Forest plot of studies comparing the risk of new onset AF in patients who had psoriasis and control. A diamond data marker represents the overall risk ratios and its 95% CI.

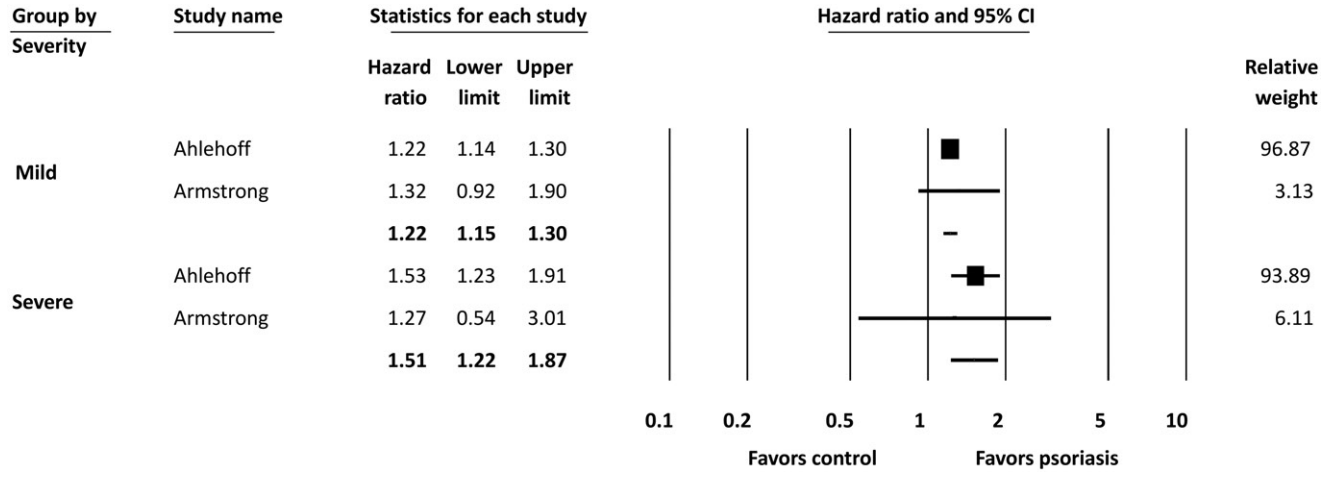

Figure 2. Forest plot of studies comparing the risk of new onset AF in patients who had bariatric surgery and those did not have bariatric surgery. Subgroup of psoriasis severity: mild and severe.

All three results were statistically significant, showing that new onset atrial fibrillation is more common in patients with psoriasis whether it be mild, moderate or severe psoriasis. These results were found to be independent of age, gender, co-morbidity, concomitant medication and socioeconomic status. These novel results add to already existing evidence of the role of psoriasis as a clinically relevant cardiovascular risk factor and they call for increased awareness of cardiovascular risk factor management in this large group of patients.

The strengths of the study lie in the large pooled cohort of study subjects including Ahlehoff et al. (13) which included the entire Danish population in their database above age 10, and Parisi et al. (16) which was also adequately powered. Follow-up time frames were sufficiently long for all studies included in the meta-analysis. Finally, we conducted subgroup analysis which displayed a consistency across the board independent of disease severity.

The major limitation of our study lies in the fact that observational studies are inherently susceptible to bias and confounding. Another limitation includes incidental atrial fibrillation occurrence in patients with concurrent psoriasis, which could impact the final data and overall results. Another important limitation was our inability to account for some important cardiovascular risk factors, e.g. smoking, lipid levels, and obesity, which in turn would affect the overall results in their unique way. Furthermore, a socioeconomically and ethnically diverse patient population was studied under one umbrella whereby each of the unique patient populations with their specific co-morbidities would affect the incidence of atrial fibrillation and psoriasis independent of each other, and thus affect the results. Misclassification of cardiovascular diseases and risk factors due to, for example, untreated hypertension, diabetes, or dyslipidemia represents another potential source of bias. In order to eliminate this potential bias, additional cohorts should be studied in specific patient populations with respect to cardiovascular risk factors, specific comorbid conditions to further investigate the association between psoriasis and atrial fibrillation. The studies in these additional populations could help address possible reasons for discordant findings if they were present. With study populations studied with new-onset atrial fibrillation, paroxysmal or persistent could possibly have missed episodes of asymptomatic paroxysmal atrial fibrillation.

Thus, our results may underestimate atrial fibrillation occurrence and primarily detect symptomatic atrial fibrillation between groups with and without psoriasis. Furthermore, we see that in our cohort a discordant number of patients was exposed to biologic treatments for psoriasis, thus it was not possible to investigate whether biologic therapy potentially reduces the risk of major CV events, including atrial fibrillation. Psoriasis has previously been associated with cardiovascular risk factors and cardiovascular morbidity and mortality (18–21). Furthermore, there are reports of decreased cardiovascular risk in patients with psoriasis receiving treatment with either systemic anti-inflammatory therapy including methotrexate or tumor necrosis factor-alpha inhibitors and, conversely, a reduction in psoriatic skin manifestations of plaque after treatment with statins has been suggested in literature (22–24).

Overall, this review showed strong evidence of association between atrial fibrillation and psoriasis irrespective of disease severity. Based on the data we found and our study results, it would be of interest to know if such a relationship exists between patients with psoriasis and concurrent atrial fibrillation, and if treatment with one would beneficially affect the other. However, it would still stand true, based on existing evidence, that patients with inflammatory arthritis are at increased risk of CVD and this may be an additional reason to minimize the patient's inflammatory burden. Another arena of further research would include delineating if other inflammatory diseases were associated with atrial fibrillation based on the existing association between the two, considering inflammation is likely to be a common denominator. Lastly, as mentioned before, it would be prudent to sub-analyze different patient populations with atrial fibrillation and psoriasis to establish a more firm causal linkage between the two inflammatory conditions.

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Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Disclosure statement

The authors declare no conflicts of interest.

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Appendix

MEDLINE

1 psoriasis.mp. or exp Psoriasis/

2 psoria*.mp. [mp = title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier]

3 1 or 2

4 Atrial Fibrillation.mp. or exp Atrial Fibrillation/

5 auricular fibrillation.tw.

6 atrium fibrillation.tw.

7 4 or 5 or 6

8 3 and 7

EMBASE

((('atrial fibrillation'/exp or 'atrial fibrillation') or (atrial or 'atrium'/exp or auricular and fibrillat*)) and (('psoriasis'/exp and [embase]/lim) or (psoria* and [embase]/lim))) and [embase]/lim not [medline]/lim

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